

An Instructor-Configurable Digital Twin for Autonomous Course Tutoring

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Abstract

This project designs and implements an instructor-configurable Digital Twin that answers course questions and initiates support between student requests. Reviewed materials, teaching preferences and objectives are connected to evidence selection, persistent learner records, action planning and in-app delivery. Comparative studies examine evidence interfaces, guarded planning, model allocation and learner-estimator/timing combinations. A typed evidence interface improved grounded answers from 63.73% to 89.42% on one development fold. On a different fresh-case evaluation, the retained BM25 fallback met the full grounding criteria in 63.25% of answerable cases, below its acceptance threshold. Integrated trials using external large language models (LLMs) exercised 24 histories over 30 virtual days, including proactive delivery, consent changes and restart. A subsequent objective-scoped life-cycle correction reduced unsupported goal completions from 12 to zero in matched 72-history arms. Mean final simulator mastery did not increase relative to the previous completion logic. The results establish a working research prototype and specific design findings; they also expose a gap between controlled execution and individualized teaching. The evaluations used synthetic students and teaching profiles; effects on real-student learning and fidelity to the actual instructor remain untested.

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